

SAURAJ BHARTI

INSPIRE-HEP

PERSONAL DATA

PLACE AND DATE OF BIRTH: Karkora, Uttar Pradesh, India | July 1995
NATIONALITY AND GENDER: Indian | Male
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PHD

TENURE : Jan 2020 - June 2026
RESEARCH TOPIC : Mock catalogues of galaxies and prospects for gravitational lensing of the redshifted 21 cm line
SUPERVISOR : [Prof. Jasjeet Singh Bagla](#)
INSTITUTE : Indian Institute of Science Education and Research (IISER), Mohali

PAST RESEARCH OVERVIEW

My research broadly centres on generating mock catalogues of galaxies for radio observations of atomic Hydrogen at intermediate redshifts and investigating galaxy evolution across multiple wavelengths. Using these mocks, I investigate the prospects for detecting redshifted 21 cm emission in forthcoming SKA surveys. The following approaches were used for detection prospects: HI stacking, photometry, and gravitational lensing.

I look forward to applying my expertise in creating mock catalogs of 21-cm emission at different epochs and to collaborating on related topics of mutual interest: hydrodynamical simulation, EoR and gravitational lensing.

PUBLICATIONS LIST

DEC 2022 : Sauraj Bharti and Jasjeet Singh Bagla,
Upcoming SKA precursor survey and sensitivity to HI mass function,
[J Astrophys Astron 43, 95 \(2022\)](#)
OCT 2025 : Sauraj Bharti, Ashish K. Meena and Jasjeet Singh Bagla,
Gravitational lensing of 21 cm HI signal: detection prospects at $z \sim 1$ with
uGMRT in galaxy cluster lenses
[MNRAS, Volume 548, Issue 1, May 2026](#)
JAN 2026 : Sauraj Bharti and Jasjeet Singh Bagla,
Joint Optical-HI catalogues and prospects for upcoming HI surveys
[MNRAS, Volume 550, Issue 2, August 2026](#)

EDUCATION

JULY 2018 Master of Science in PHYSICS,
Department of Physics, Indian Institute of Technology, Madras
Major: Cosmology | CGPA: 6.91/10

JULY 2016 Bachelor of Science in PHYSICS
DPBS PG College, Anoopshahr, **CCS University** | PERCENTAGE: 72.5%

JULY 2013 Higher Secondary Certificate
Kaynat Girls Inter College, Rajpura, **UP State Board** | PERCENTAGE: 86.4%

JULY 2011 Secondary School Certificate
Suraj Mukhi Inter college, Gowan, **UP State Board** | PERCENTAGE: 70.5%

OTHER RESEARCH EXPERIENCE

JULY 2017 - JULY 2018 **Masters project at IIT, Madras, India**

Magnetic characterization of BaFeO₃ based material and analysis

Supervisor: Prof. K. Sethupati, IIT, Madras, India.

CONFERENCES, SCHOOLS AND WORKSHOPS

- JUNE 2026: (Workshop: 8–25 June)
 Physics of Galaxy Formation
 IUCAA, Pune, India
 Completed hands-on training in MP-Gadget (DM+Hydrodynamical), along with tutorials on EoR and radiative transfer methods.
- JAN 2026: (presented talk + poster)
 SKA India Science Meeting
 IISER Mohali, India
 Joint Optical-HI catalogues and prospects for upcoming HI surveys
- SEPT 2025: (presented talk)
 Challenges and Innovations in Computational Astrophysics VI
 IISER Mohali, India
 Joint Optical-HI catalogues and prospects for upcoming HI surveys
- OCT 2024: (presented talk)
 11th KIAS Workshop on Cosmology and Structure Formation
 Held at Gyeongju, South Korea
 HI-stacking prediction for upcoming surveys with SKA precursors
- SEP 2024: (presented poster+talk)
 7th Global 21cm workshop
 Held at Raman Research Institute, Bangluru, India
 HI-stacking prediction for upcoming surveys with SKA precursors
- MAY 2024: (presented talk)
 Spectrum of Physics: Particle to Universe
 International e-conference organised by Lajpat Rai College, Sahibabad
 Talk title: HI-stacking prediction for upcoming surveys with SKA precursors
- FEB 2024: (presented poster talk)
 MeerKAT@5 Meeting
 Held at STIAS Stellenbosch, South Africa
 HI-stacking prediction for upcoming surveys with SKA precursors(MeerKAT only)
- NOV 2023: (presented talk)
 Challenges and Innovations in Computational Astrophysics V
 Virtual Meeting
 Talk title: HI-stacking prediction for upcoming surveys with SKA precursors
- DEC 2023: (presented poster talk)
 International Conference on Gravitation and Cosmology 2023
 Held at IIT Guwahati, India
 HI-stacking prediction for upcoming surveys with SKA precursors
- JAN 2023: (presented talk)
 7th Shivalik HEPCATS Meeting
 Held at IIT Mandi, Himachal Pradesh, India
 Talk title : HI and optical correlation of galaxies and their mock surveys with MeerKAT

DEC 2022:	(presented talk) Meeting on "Gas in Galaxies" Held at Evolve Back Resort, Coorg, Karnataka: Organised by the Indian Academy of Science and RRI Bangalore Talk title : Upcoming SKA precursors surveys in HI-stacking and sensitivity to HI mass function
MARCH 2022:	(presented poster) 40th Meeting of Astronomical Society of India Held at IIT Roorkee, India Poster title: Upcoming SKA precursors surveys and sensitivity to HI mass function
MAY 2021:	Workshop and School on 21 cm Cosmology and Re-ionization Two days e-workshop with hands-on session on Radio Astronomy
DEC-JAN 2017:	(presented talk) Sakura Science Program and Robotics Workshop 12 days Science Workshop: held at Sibaura Institute of Technology, Tokyo, Japan Talk title: Culture, Science and Heritage sites of Japan (Asakusa Temple)
DEC 2016:	Astronomy and Astrophysics Winter School Held at IIST, Trivandrum, Kerala 10 days School on basics of Astronomy: Project: Astronomy source catalog selection

FELLOWSHIPS, SCHOLARSHIPS AND AWARDS

BEST PRESENTATION PRIZE 2024	Spectrum of Physics: Particle to universe, International e-conference organised by LR College, Sahibabad Certificate of appreciation in Astrophysics and Cosmology
GATE PHYSICS 2024:	All India Rank- 2600
GATE PHYSICS 2021:	All India Rank- 1650
GATE PHYSICS 2019:	All India Rank- 1823
UGC NET-JRF JUNE 2019:	For lectureship in physics, AIR 244
JAM PHYSICS 2016:	All India Rank- 1545
INSPIRE SHE:	INSPIRE Scholarship for Higher Education by DST India Tenure: 2013-2018

LANGUAGES

MOTHER TONGUE:	Hindi
FLUENT:	English, Hindi

TEACHING AND OTHER ACTIVITIES

DEC 2022:	Volunteer in XXV DAE-BRNS HEP Symposium
AUG-NOV 2022:	TA for PHY411 Nuclear Physics Lab (IISER Mohali)
FEB-MAY 2022:	TA for PHY111 Mechanics Lab (IISER Mohali)

Apart from research, I value teaching and outreach. During the mid-time of my PhD at IISER Mohali, I served as a teaching assistant for several undergraduate and master's physics courses and also mentored students in computational projects.

COMPUTER SKILLS

HIGH PERFORMANCE COMPUTING :	PARAMSMRITI(NABI) , HPC-IISERM
PROGRAMMING LANGUAGES :	Python, Fortran90, C, C++.
OPERATING SYSTEMS :	Linux based systems
RADIO ASTRONOMY, IMAGING AND FLAGGING :	CASA , DS9